A Dual-Tier Edge-Cloud Architecture for Real-Time Basketball Activity Recognition and Fatigue Monitoring.

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert.**

I. Introduction

The constant pursuit of efficiency and durability extends not only to technology but also to human performance. Central to this evolution is the rapid progression of wearable technology, which has transitioned from simple step-counting peripherals to sophisticated diagnostic tools. These devices now encompass a vast range of applications, including medical diagnostics, injury prevention, sleep analysis, and high-fidelity sports performance monitoring.

In recent years, the synergy between advanced microcontrollers and smartphone has revolutionized the sports landscape. Modern athletics is increasingly defined by data-driven decision-making, where statistics are utilized to gain competitive advantages. However, as the demand for deeper insights grows, the complexity of the data required to satisfy these metrics has increased proportionally.

Compared to single-activity recognition, team sports present significant challenges for real-time analytics. The inherent unpredictability, high-intensity intervals, and constant physical interaction between players create a volatile data environment. These sports are characterized by explosive, short-duration movements and high levels of physiological stress, especially during a game's climax where the risk of injury is most prevalent [1].

Despite these advancements, standard sports wearables remain limited by their reliance on single-modality sensing, typically utilizing only Inertial Measurement Units (IMU). While kinematics provide a trajectory of movement, they lack the neuromuscular context required to differentiate between explosive intent and mere mechanical motion. Furthermore, the reliance on cloud-based processing introduces latency that can be critical in injury-prevention scenarios.

This paper addresses these gaps by proposing a multi-modal, dual-tier architecture that fuses IMU kinematics with Electromyography (EMG). By leveraging Edge AI for localized inference on a mobile hub, the system provides a low-latency solution for movement classification and fatigue monitoring, ensuring that the physiological cost of movement is captured as accurately as its physical path.

II. Related Work

The development of intelligent sports wearables has increasingly focused on objective physiological monitoring to supplement traditional kinematic tracking. Recent research demonstrates a paradigm shift toward multi-modal sensor fusion and edge-based analytics to enhance the accuracy of activity recognition and fatigue detection.

A common challenge in athletic monitoring is the inability of single-modality sensors to provide full biomechanical context. Hwang et al. addressed this by integrating surface electromyography (sEMG) and inertial measurement units (IMU) to classify physical fatigue during natural over-ground walking. Their hybrid CNN-LSTM-Attention model, evaluated via leave-one-subject-out cross-validation, achieved 87.94% accuracy, proving that combining physiological markers from the gastrocnemius with kinematic jerk signals significantly enhances system robustness against individual variability. Similarly, Otálora et al. demonstrated that for upper limb fatigue estimation during repetitive lifting, a combination of IMUs and Optical Fiber Sensors (OFS) achieved 95.4% accuracy. Their findings highlight that while EMG is a gold standard for bio-electrical muscle function, its reliability in long-term work environments can be compromised by factors like skin perspiration, making supplemental kinematic data essential.

The choice of classifier and processing pipeline is critical for real-time mobile deployment. Baigarayeva et al. developed an IoT-based system, "Qimyl," which performs on-device feature extraction (RMS, MAV, MDF) to classify muscle fatigue during sustained isometric tasks. By comparing logistic regression, decision trees (DT), and XGBoost, they identified DT as achieving the highest accuracy (90.7%) for their specific submaximal grip tasks. Furthermore, their work emphasizes the value of synchronizing objective EMG features with sub-

jective self-assessments (e.g., the Borg scale) to improve the ecological validity of the models.

Beyond standard classification, researchers have explored deterministic indices to objectively detect the moment of failure. Qassim et al. proposed a double-step binary classifier utilizing a new fatigue index based on the difference between high-frequency and low-frequency components of the sEMG signal. This algorithm achieved 94.66% accuracy in identifying the moment a muscle fails to sustain required force, providing a blueprint for active injury-prevention alerts. These findings are supported by Kamaruddin et al., whose review of surface electromyography emphasizes that while time-domain features like Waveform Length (WL) offer low computational complexity, frequency-domain analysis remains superior for quantifying the physiological shift to lower frequencies that characterizes neuromuscular fatigue.

III. System Architecture

The proposed system follows a decentralized, multi-modal architecture designed to bridge localized physiological sensing with high-level cloud analytics. The framework is divided into three primary layers: the sensor-embedded hardware, the mobile edge-computing hub, and the cloud-based inference server.

A. Hardware Prototype and Data Acquisition

The sensory layer consists of two footwear-mounted prototypes. Each unit is centered around an Arduino Nano 33 BLE Sense Rev2, chosen for its embedded 9-axis Inertial Measurement Unit (IMU) and Bluetooth Low Energy (BLE) module. To provide physiological context, a localized Electromyography (EMG) sensor is integrated into each module to monitor the electrical activity of the Gastrocnemius Lateralis.

The firmware implements a non-blocking sampling strategy. EMG signals are polled at 1000 Hz and subjected to a 20 Hz high-pass filter to remove baseline drift. Six time-domain features (RMS, MAV, VAR, ZC, SSC, and WL) are calculated every 200 ms. Simultaneously, IMU kinematics are sampled at 50 Hz. To optimize BLE bandwidth, the system bundles five 30-byte frames into a single 150-byte payload, transmitted to the mobile hub every 100 ms.

B. Data Pipeline and Temporal Smoothing

The Android-based mobile application acts as the User Interface (UI), implemented using a Model-View-ViewModel (MVVM) architecture. Upon receiving the bilateral BLE data, the software synchronizes the left and right insole data into a unified 30-feature vector per timestamp.

To handle the inherent noise of dynamic basketball movements, the system utilizes a 300 ms sliding window with a 100 ms step size. A critical contribution of this architecture is the implementation of a post-processing temporal smoothing layer. Utilizing a continuous sliding window (ArrayDeque), a Majority Voting algorithm evaluates the last five predictions (500 ms context). This logic acts as a digital low-pass filter, effectively eliminating prediction flickering and hardware-induced false positives during steady-state athletic movements.

C. Dual-Tier Machine Learning Inference

The architecture employs a dual-tier approach to compare latency and model complexity:

- Edge Tier (TinyML): A lightweight, quantized Neural Network (NN) is embedded directly within the Android application using TensorFlow Lite. This model provides ultra-low latency inference, enabling real-time feedback and driving the deterministic injury-prevention algorithm independently of network availability.
- Cloud Tier (Random Forest): Simultaneously, the 300 ms window is serialized into a JSON payload and transmitted via an asynchronous HTTP POST request to a Flask-based server. The server hosts a high-dimensional Random Forest (RF) classifier used for deep analysis and comparative validation against the edge model.

IV. Experimental Method and Results

A. Experimental Method

The system was evaluated through a series of dynamic basketball drills involving several volunteer subjects. The dataset was partitioned into an 80/20 train-test split to validate the classification models across five macro-movements: Idle, Walking, Running, Defensive Slide, and Rebound. The cramping prevention system was only tested on one player that pushed his muscles to near involuntary contractions.

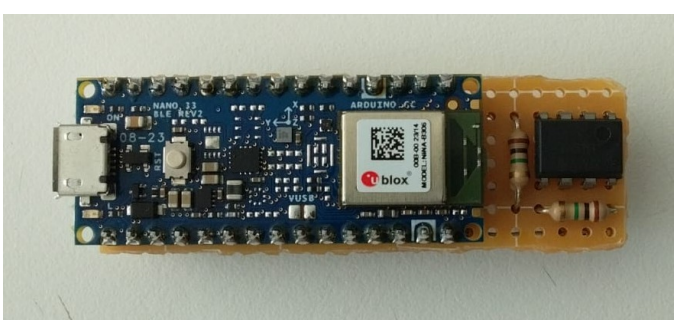

Fig. 1. Prototype Picture.

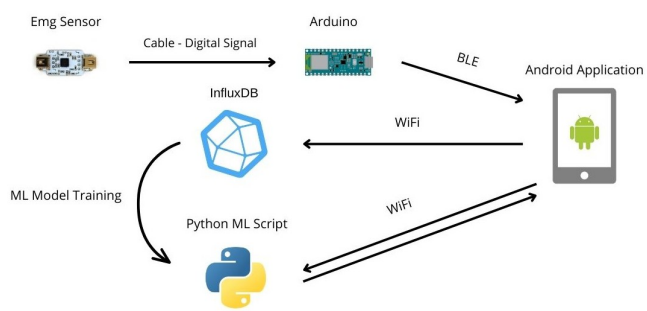

Fig. 2. System Overview.

The prototypes were mounted directly in the athletes' footwear to approximate the ideal positioning(on the bottom of the feet as an insole or a sock). The devices were secured using the shoe laces and the powerbank was secured in the users socks, utilizing the existing infrastructure. With this configuration, the IMU accurately captures the kinematics of the foot, such as accelerations and rotational velocities during the athletes' movements. Furthermore, this setup minimizes the mechanical strain exerted by the power cables on the USB ports, while keeping the EMG leads in close proximity to the chosen muscle group.

B. Model Training and Convergence

The TinyML model demonstrated robust learning, achieving a validation accuracy of approximately 94high generalization. Conversely, the Cloud-based Random Forest (RF) was evaluated using ROC and Precision-Recall curves. The Area Under the Curve (AUC) for all classes approached 1.0, and the PR curves confirmed the model's ability to handle the class imbalance inherent in basketball data (e.g., rare rebounding events vs. frequent running/walking).

C. Feature Importance and Sensor Contribution

A core contribution of this work is the validation of EMG integration. Gini Feature Importance analysis (3) revealed that EMG-derived features, specifically Variance (VAR) and Root Mean Square (RMS), were the primary contributors to classification accuracy. This confirms that physiological electrical activity provides critical intent-based context that kinematics alone lack. Additionally, the model exhibited a preference for features from the dominant (left) limb, successfully capturing subject-specific biomechanical asymmetries.

D. Edge vs. Cloud: Real-Time Performance

During live testing, the localized TinyML model outperformed the Cloud RF model, achieving an overall accuracy of 79.5% compared to the server's 71.2%. While the RF model excelled at slower movements (Idle, Walking), it struggled with dynamic transitions in the field. Notably, the TinyML model provided instantaneous inference, whereas the RF model introduced measurable network latency.

E. Temporal Smoothing and Stability

The 500ms majority-voting algorithm provided a significant stability boost. For the TinyML model, smoothing increased overall accuracy by 6.4% by acting as a digital low-pass filter that neutralized prediction flickering. Confusion matrix analysis (4, 5) shows that while smoothing improved steady-state detection, it reduced sensitivity to transient events like "Rebounding," which were often outvoted by surrounding states. This identifies a fundamental trade-off: smoothing is essential for consistent activity tracking but may require dynamic windowing for explosive event detection in future iterations.

V. Conclusion

In summary, the localized TinyML model outperformed the Cloud-tier RF (79.5% vs 71.2% accuracy), while providing the instantaneous feedback required for injury prevention. The implementation of a 500ms majority-voting algorithm successfully improved classifications despite the identified trade-off in sensitivity regarding explosive movements like rebounds.

Future work will focus on transitioning the prototype into a fully integrated smart-textile garment to improve ergonomics and signal stability. Additionally, expanding the dataset demographic and correlating EMG threshold breaches with clinical physiological markers will further refine the system's predictive accuracy and universal applicability.

References

[1] Lutz, Jonas & Memmert, Daniel & Raabe, Dominik & Dornberger, Rolf & Donath, Lars. (2019). Wearables for Integrative Performance and Tactic Analyses: Opportunities, Challenges, and Future Directions. International Journal of Environmental Research and Public Health. 17. 59. 10.3390/ijerph17010059.

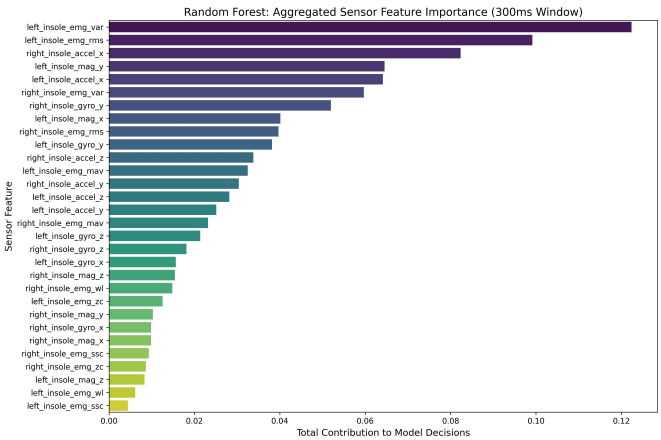

Fig. 3. Feature Importance.

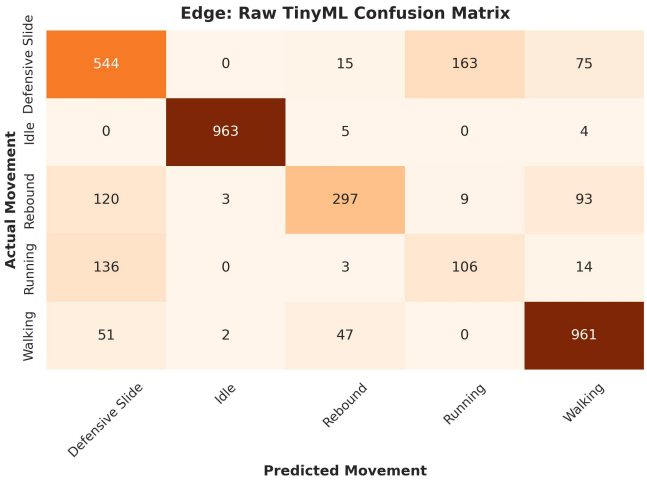

Fig. 4. TinyML Confusion Matrix.

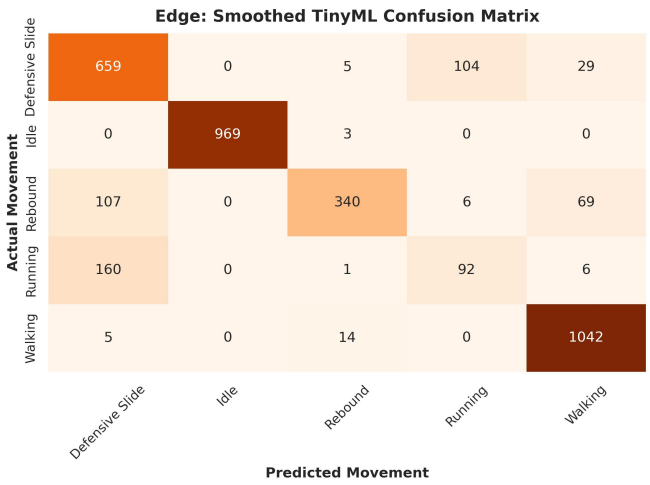

Fig. 5. TinyML with Smoothing Confusion Matrix.